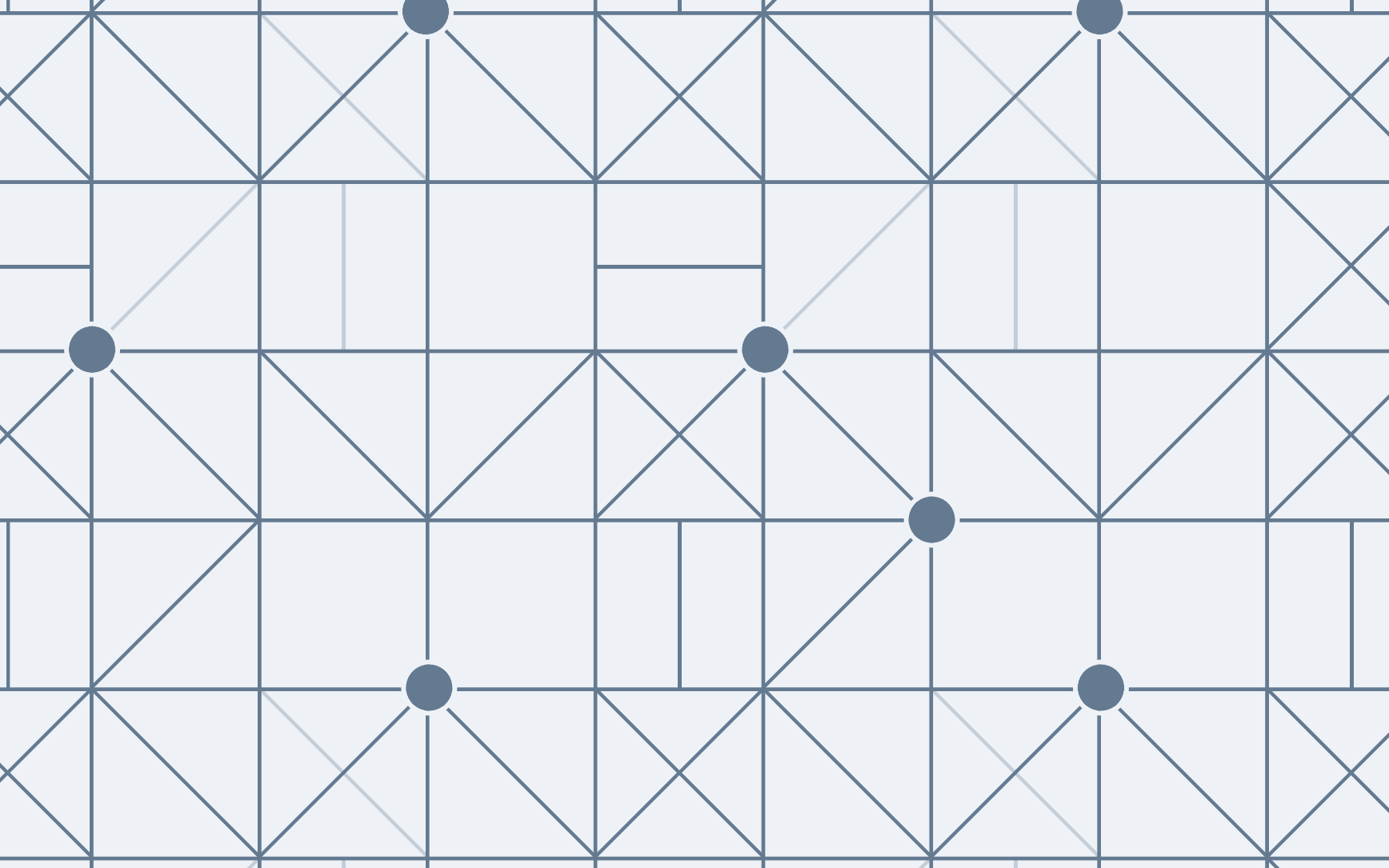




Short-Run : $SR TC(q) = w L^{SR}(q) + r K_0$

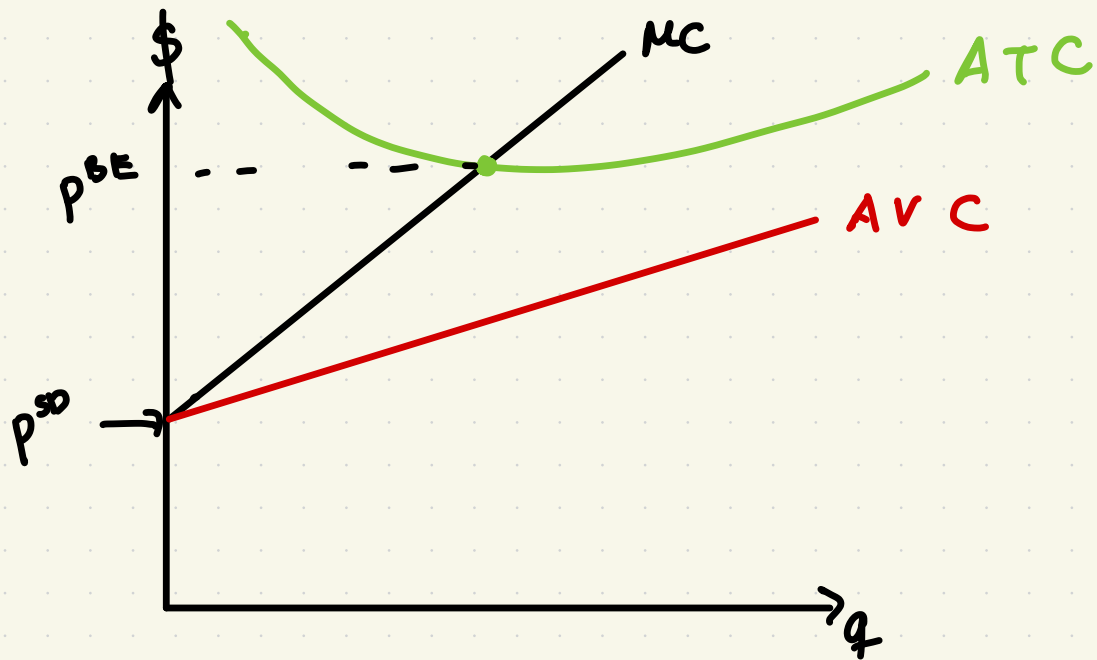
(at least
one-input
is fixed
↳ sunk)

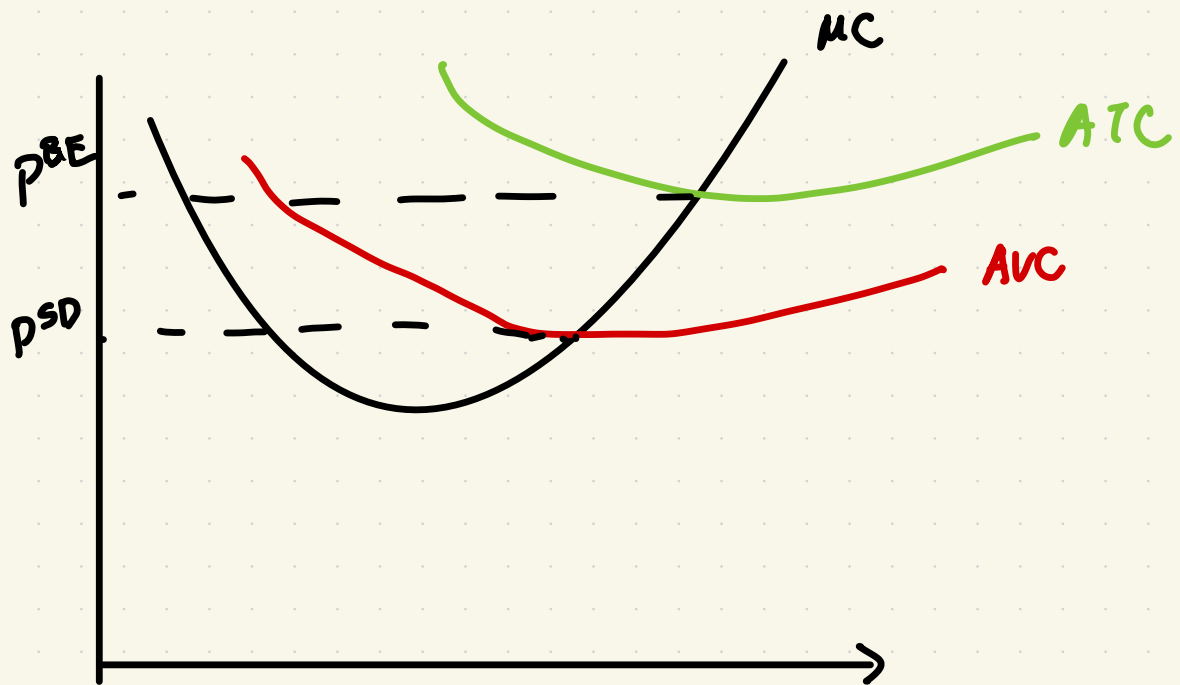
$$\hookrightarrow \frac{\Delta SR TC}{\Delta q} = \frac{d SR TC}{dq} = MC(q)$$

Long-Run : $LR TC(q) = w L^*(q) + r K^*(q)$

(Solve full
cost min
problem
of the firm
no-inputs fixed
no-sunk costs)

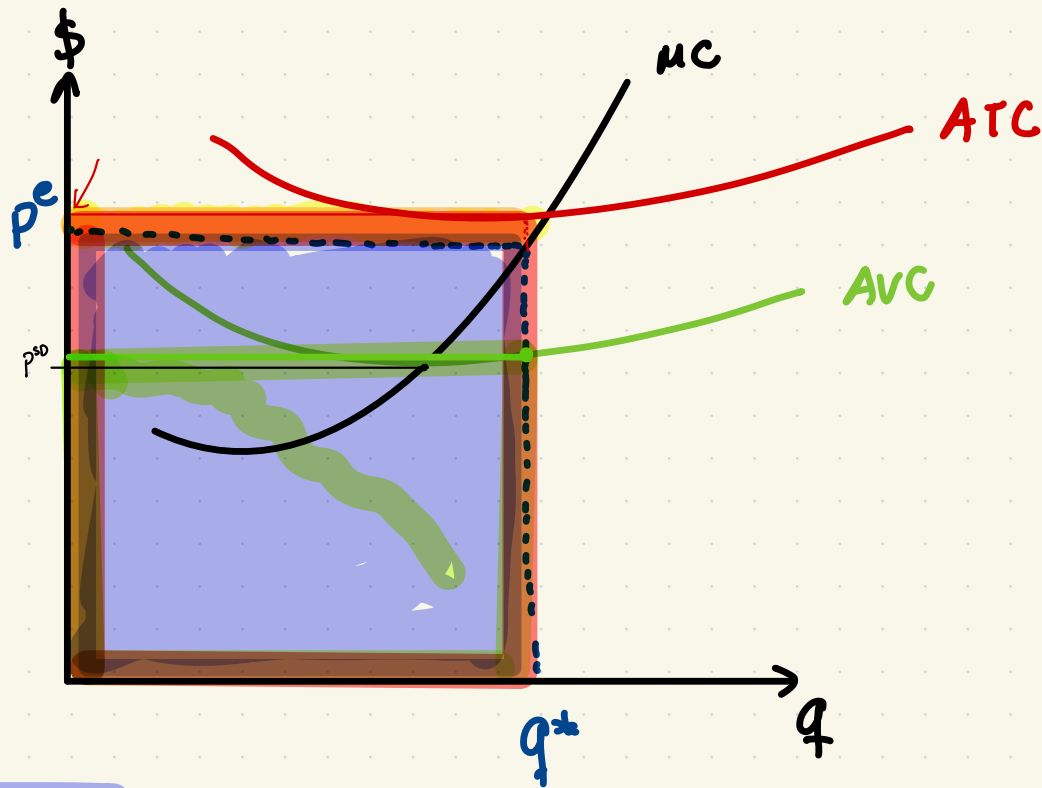
$$\hookrightarrow \frac{\Delta LR TC(q)}{\Delta q} = \frac{d LR TC(q)}{dq} = MC(q)$$





$$A_\nu f(x) = \frac{f(x)}{x}$$

$$x \cdot A_\nu f(x) = f(x)$$

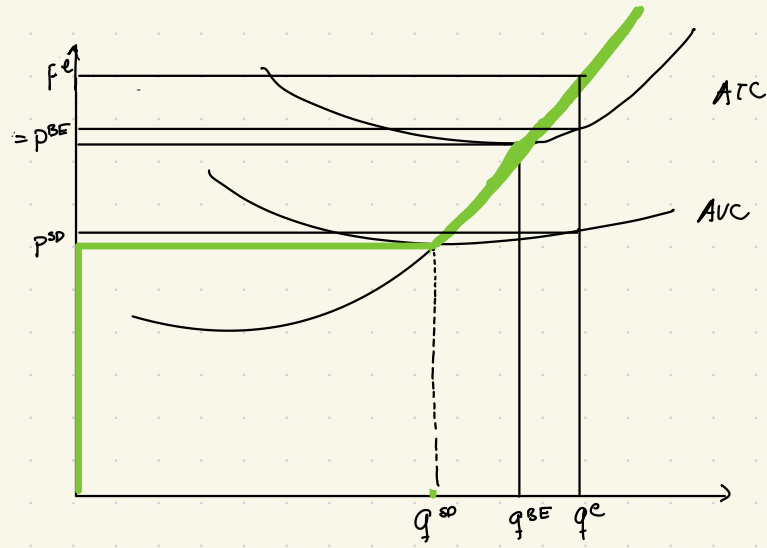


$$\text{Revenue} = p^e \cdot q^+$$

$$TC = q^+ ATC(q^+)$$

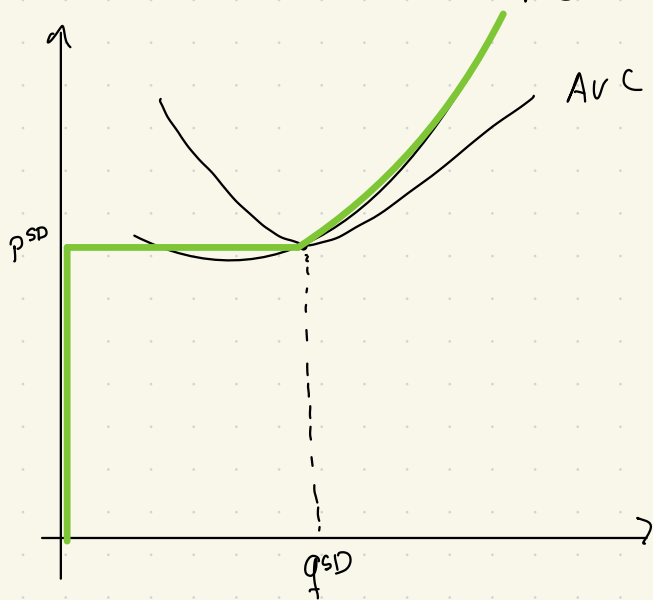
$$VC = q^+ AVC(q^+)$$

Losses



— Firm Supply

Short-Run Supply



$$\text{Supply } p = MC(q) \quad p \geq p^{SD}$$

otherwise

invert

$$q_i^s = q_i^s(p)$$

with identical firms :

$$Q^s(p) = \sum_{i=1}^N q_i^s(p) = N q^s(p)$$

Q^D aggregate demand

Q^s aggregate supply

$$Q^D(p^*) = Q^s(p^*) = Q^*$$

p^* is the price the firm will take as given

$$p^* = MC(q^*)$$

$$q^* = Q^*/N$$

$$STC(q) = q^2 + 10$$

$$MC(q) = 2q$$

$$AVC(q) = q$$

$$APC = 10/q$$

$$p^{\text{sd}} = 0$$

$$p = MC(q) = 2q \quad | \quad p \geq 0$$

$$q^s = P/2$$

$$N \text{ identical firms: } Q^s(p) = \frac{N}{2} p$$

$$Q^D(p) = 100 - 8p$$

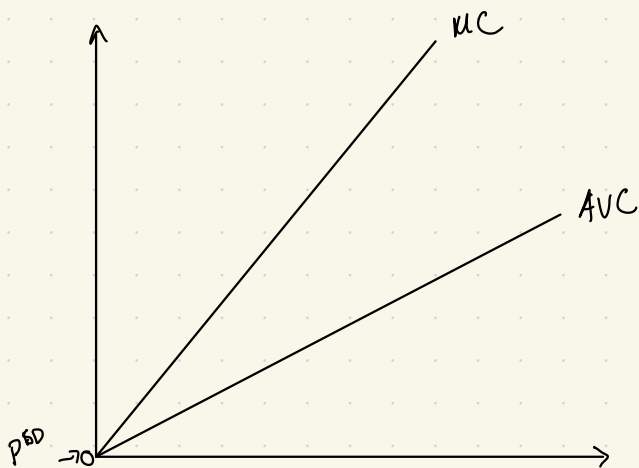
$$p^* \text{ solves: } \frac{N}{2} p^* = 100 - 8p^*$$

$$\Rightarrow \left(\frac{N}{2} + 8 \right) p^* = 100$$

$$p^* = \frac{100}{N/2 + 8} = \frac{200}{N + 16}$$

$$Q^* = \frac{N}{2} \left(\frac{200}{N + 16} \right) = \frac{100N}{N + 16}$$

$$q^* = \frac{200}{N + 16} \cdot \frac{1}{2} = \frac{100}{N + 16}$$

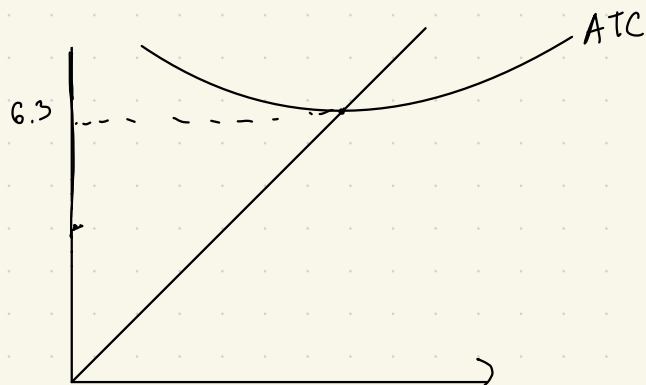


$$LR TC(q) = \begin{cases} q^2 + 10 & q > 0 \\ 0 & q = 0 \end{cases}$$

$$MC(q) = 2q \quad ATC(q) = q + 10/q$$

$$AVC(q) = q$$

$$AFC(q) = 10/q$$



$$p^0 = \min ATC$$

$$\hookrightarrow p^{BE}: MC = ATC$$

$$2q^0 = q^0 + \frac{10}{q^0}$$

$$\Rightarrow 2q^{0^2} = q^{0^2} + 10$$

$$q^{0^2} = 10$$

$$q^0 = \sqrt{10}$$

$$p^0 = 2\sqrt{10}$$

$$\text{Recall } Q^D = 100 - 8p$$

$$Q^{LR} = 100 - 8p^0 = 100 - 16\sqrt{10}$$

$$\approx 49.4$$

$$N^{LR} = \left\lfloor \frac{Q^{LR}}{q^0} \right\rfloor = \left\lfloor \frac{100 - 16\sqrt{10}}{\sqrt{10}} \right\rfloor$$

Floor function.